

01_hypertuner

October 4, 2025

```
[1]: import time
import numpy as np
from plotly import graph_objects as go
import matplotlib.pyplot as plt

from loguru import logger

import ray
from ray import tune

import torch
from torch import optim
from torch import nn

from pathlib import Path

%matplotlib inline
plt.style.use('dark_background')
# import plotly.io as pio
# pio.renderers.default = 'plotly_mimetype+notebook'

import visualize
DELETE = True # to delete the tunedir at the end of the notebook
start = time.time()
```

This is a general reference notebook to explore the use of ray tuner. First, we define some global variables. This makes it easier to change the parameters for the full notebook at once and run everything.

Set the max_epochs to something like 10 (this will take longer, but you will get better results), and experiments to 18. This way, you will have a good way to compare gridsearch with the other tuners because gridsearch will do $3 * 6 = 18$ experiments.

The reason I am setting this low is because i automated the testing of the notebook and it takes a long time to run all the experiments by default.

```
[2]: MAX_EPOCHS = 10
N_EXPERIMENTS = 18
```

1 Train function and config

Let's define some dicts to log the results of the training.

```
[3]: timer = {}  
     best_config = {}
```

We will need a training function. This is also implemented in the mltrainer, but I put it here to show you how the details work.

```
[4]: def train_fn(model, trainstreamer, lossfn, optimizer, steps):  
     model.train()  
     train_loss: float = 0.0  
     for _ in range(steps):  
         x, y = next(trainstreamer)  
         optimizer.zero_grad()  
         yhat = model(x)  
         loss = lossfn(yhat, y)  
         loss.backward()  
         optimizer.step()  
         train_loss += loss.item()  
     return train_loss
```

We validate on the validation set and return the loss.

```
[5]: def validate(model, validstreamer, lossfn, metric, steps):  
     model.eval()  
     valid_loss: float = 0.0  
     acc: float = 0.0  
     with torch.no_grad():  
         for _ in range(steps):  
             x, y = next(validstreamer)  
             yhat = model(x)  
             loss = lossfn(yhat, y)  
             valid_loss += loss.item()  
             acc += metric(y, yhat)  
     acc /= steps  
     return valid_loss, acc
```

Getting the data requires a bit extra care. Because we will run experiments in parallel on all the cpu's available, we well use FileLock to make sure that loading the data does not conflict.

Note we import functions inside the function we will later on pass to ray. This is because ray will serialize the function and send it to the workers. If we import the functions outside the function, the workers will not have access to them.

```
[6]: def get_data(tune_dir):  
     from filelock import FileLock  
     from mads_datasets import DatasetFactoryProvider, DatasetType  
     from mltrainer.preprocessors import PaddedPreprocessor
```

```

from mads_datasets import DatasetFactoryProvider, DatasetType
with FileLock(tune_dir / ".lock"):
    # we lock the datadir to avoid parallel instances trying to
    # access the datadir
    preprocessor = PaddedPreprocessor()
    gesturesdatasetfactory = DatasetFactoryProvider.
↪create_factory(DatasetType.GESTURES)
    streamers = gesturesdatasetfactory.create_datastreamer(
        batchsize=32, preprocessor=preprocessor
    )
    train = streamers["train"]
    valid = streamers["valid"]
    return train, valid

```

We will use the same GRU model we used last lesson. You might improve this in a few ways (eg consider adding skip layers, conv1d, etc) but for clarity, lets keep it simple.

```

[7]: class GRUmodel(nn.Module):
    def __init__(
        self,
        config: dict,
    ) -> None:
        super().__init__()
        self.rnn = nn.GRU(
            input_size=config["input_size"],
            hidden_size=int(config["hidden_size"]),
            dropout=config["dropout"],
            batch_first=True,
            num_layers=int(config["num_layers"]),
        )
        self.linear = nn.Linear(int(config["hidden_size"]), ↵
↪config["output_size"])

    def forward(self, x):
        x, _ = self.rnn(x)
        last_step = x[:, -1, :]
        yhat = self.linear(last_step)
        return yhat

```

Now, we have all the ingredients we need to run the tuner. We create a function that: - loads the data with a lock - creates the model - trains the model - validates the model - reports the results to ray

```

[8]: def tune_model(config: dict):
    from mltrainer.metrics import Accuracy

    # load data
    train, valid = get_data(config["tune_dir"])

```

```

trainsteps = len(train)
validsteps = len(valid)
trainstreamer = train.stream()
validstreamer = valid.stream()

# create model with config
model = GRUModel(config)
optimizer = optim.Adam(model.parameters(), lr=0.001)
loss_fn = torch.nn.CrossEntropyLoss()
metric = Accuracy()

for _ in range(config["epochs"]):
    # train and validate
    train_loss = train_fn(model, trainstreamer, loss_fn, optimizer,
↪trainsteps)
    valid_loss, accuracy = validate(model, validstreamer, loss_fn, metric,
↪validsteps)

    # report to ray
    ray.train.report({
        "valid_loss": valid_loss / validsteps,
        "train_loss": train_loss / trainsteps,
        "accuracy" : accuracy,
    })

```

Let's try this, to see if everything works as expected. Note that it typically does not make sense to use acceleration here; we will typically have 1 GPU, but 10 or 16 CPUS. We gain much more from parallelizing the experiments than we gain by running the model faster on a single GPU.

```
[9]: tune_dir = Path("logs/test/").resolve()
```

```
[10]: config = {
    "input_size": 3,
    "output_size": 20,
    "dropout": 0.1,
    "hidden_size": 64,
    "num_layers": 2,
    "epochs": MAX_EPOCHS,
    "tune_dir": tune_dir,
}
tune_model(config)
```

```
2025-10-01 22:09:04.524 | INFO      |
```

```
mads_datasets.base:download_data:121 - Folder
```

```
already exists at /Users/nickreinders/.cache/mads_datasets/gestures
```

```
100%|      | 2600/2600 [00:00<00:00, 5312.90it/s]
```

```
100%|      | 651/651 [00:00<00:00, 5689.41it/s]
```

```
/Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deploymen
```

```
t/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-
packages/ray/train/_internal/session.py:657: UserWarning: `report` is meant to
only be called inside a function that is executed by a Tuner or Trainer.
Returning `None`.
    warnings.warn(
```

2 Random search

Now, let's do a random search.

First, we define the search space. We can specify specific values, but also distributions. For the hidden size, we will use `randint`, which will sample from a uniform distribution of integers. The same for the number of layers.

The `tune.run` function runs the hypertuning. It will sample from the search space, and create a specific config for each experiment.

```
[11]: config = {
        "input_size": 3,
        "output_size": 20,
        "dropout": 0.05,
        "epochs": MAX_EPOCHS,
        "hidden_size": tune.randint(16, 512),
        "num_layers": tune.randint(1, 8),
        "tune_dir": tune_dir,
    }
```

Let's test this!

```
[ ]: tic = time.time()
analysis = tune.run(
    tune_model,
    config=config,
    metric="valid_loss",
    mode="min",
    storage_path=str(tune_dir),
    num_samples=N_EXPERIMENTS,
    stop={"training_iteration": MAX_EPOCHS},
    verbose=1,
)

timer["ray_random"] = time.time() - tic
```

<IPython.core.display.HTML object>

```
(tune_model pid=29281) 2025-10-01 22:09:35.511 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
 0%|          | 0/2600 [00:00<?, ?it/s]
17%|          | 449/2600 [00:00<00:00, 4484.31it/s]
```

```

39%|          | 1010/2600 [00:00<00:00, 5145.96it/s]
61%|          | 1585/2600 [00:00<00:00, 5420.53it/s]
83%|          | 2146/2600 [00:00<00:00, 5494.31it/s]
100%|         | 2600/2600 [00:00<00:00, 5382.17it/s]
 0%|          | 0/651 [00:00<?, ?it/s]
100%|         | 651/651 [00:00<00:00, 5437.96it/s]
(tune_model pid=29281) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/torch/nn/modules/rnn.py:123:
UserWarning: dropout option adds dropout after all but last recurrent layer, so non-zero dropout expects num_layers greater than 1, but got dropout=0.05 and num_layers=1
(tune_model pid=29281) warnings.warn(
47%|          | 1211/2600 [00:00<00:00, 12096.49it/s]
100%|         | 2600/2600 [00:00<00:00, 11947.78it/s]
41%|          | 1071/2600 [00:00<00:00, 10706.67it/s]
100%|         | 2600/2600 [00:00<00:00, 11169.15it/s]
100%|         | 651/651 [00:00<00:00, 11157.39it/s]
72%|          | 1875/2600 [00:00<00:00, 9489.23it/s]
100%|         | 2600/2600 [00:00<00:00, 9385.98it/s]
100%|         | 651/651 [00:00<00:00, 8503.00it/s]
30%|          | 770/2600 [00:00<00:00, 7696.92it/s]
59%|          | 1540/2600 [00:00<00:00, 7657.21it/s]
100%|         | 2600/2600 [00:00<00:00, 7351.99it/s]
100%|         | 651/651 [00:00<00:00, 7326.89it/s]
25%|          | 640/2600 [00:00<00:00, 6397.72it/s]
50%|          | 1305/2600 [00:00<00:00, 6543.83it/s]
100%|         | 2600/2600 [00:00<00:00, 6600.29it/s]
100%|         | 651/651 [00:00<00:00, 6160.04it/s]
26%|          | 679/2600 [00:00<00:00, 6787.68it/s]
56%|          | 1452/2600 [00:00<00:00, 7341.14it/s]
(tune_model pid=29281) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to `ray.tune.report` when running in a function passed to Ray Tune. This will be an error in the future. See this issue for more context: https://github.com/ray-project/ray/issues/49454
(tune_model pid=29281) _log_deprecation_warning(
84%|          | 2187/2600 [00:00<00:00, 6409.11it/s]
100%|         | 2600/2600 [00:00<00:00, 6489.63it/s]
28%|          | 729/2600 [00:00<00:00, 7270.73it/s]
56%|          | 1457/2600 [00:00<00:00, 6058.94it/s]
80%|          | 2076/2600 [00:00<00:00, 5935.29it/s]
100%|         | 2600/2600 [00:00<00:00, 6073.97it/s]
100%|         | 651/651 [00:00<00:00, 7006.92it/s]
15%|          | 399/2600 [00:00<00:00, 3986.34it/s]
35%|          | 901/2600 [00:00<00:00, 4592.49it/s]

```

```

55%|          | 1418/2600 [00:00<00:00, 4854.43it/s]
74%|          | 1913/2600 [00:00<00:00, 4890.84it/s]
(tune_model pid=29284) 2025-10-01 22:09:40.033 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 8x across
cluster] (Ray deduplicates logs by default. Set RAY_DEDUP_LOGS=0 to disable log
deduplication, or see https://docs.ray.io/en/master/ray-observability/user-
guides/configure-logging.html#log-deduplication for more options.)
100%|         | 2600/2600 [00:00<00:00, 4721.99it/s]
  0%|          |  0/651 [00:00<?, ?it/s] [repeated 16x
across cluster]
100%|         | 651/651 [00:00<00:00, 5164.78it/s]
 28%|          | 722/2600 [00:00<00:00, 7208.77it/s]
 56%|          | 1443/2600 [00:00<00:00, 5731.16it/s]
 79%|          | 2052/2600 [00:00<00:00, 5874.28it/s]
(tune_model pid=29284) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/torch/nn/modules/rnn.py:123:
UserWarning: dropout option adds dropout after all but last recurrent layer, so
non-zero dropout expects num_layers greater than 1, but got dropout=0.05 and
num_layers=1 [repeated 2x across cluster]
(tune_model pid=29284)   warnings.warn( [repeated 2x across
cluster]
100%|         | 2600/2600 [00:00<00:00, 5975.89it/s]
100%|         | 651/651 [00:00<00:00, 6146.85it/s]
(tune_model pid=29282) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 5x across cluster]
(tune_model pid=29282)   _log_deprecation_warning( [repeated 5x
across cluster]
(tune_model pid=29286) 2025-10-01 22:09:40.865 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
  0%|          |  0/651 [00:00<?, ?it/s] [repeated 2x
across cluster]
(tune_model pid=29278) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an

```

```

error in the future. See this issue for more context: https://github.com/ray-project/ray/issues/49454 [repeated 3x across cluster]
(tune_model pid=29278) _log_deprecation_warning( [repeated 3x
across cluster]
(tune_model pid=29285)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=29285) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=29286) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-project/ray/issues/49454
(tune_model pid=29286) _log_deprecation_warning(
(tune_model pid=29331) 2025-10-01 22:10:00.789 | INFO |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
0%|          | 0/2600 [00:00<?, ?it/s]
13%|         | 348/2600 [00:00<00:00, 3478.33it/s]
27%|         | 696/2600 [00:00<00:00, 2821.91it/s]
38%|         | 986/2600 [00:00<00:00, 2660.53it/s]
52%|         | 1361/2600 [00:00<00:00, 3040.55it/s]
64%|         | 1673/2600 [00:00<00:00, 2697.59it/s]
75%|         | 1952/2600 [00:00<00:00, 2516.31it/s]
88%|         | 2300/2600 [00:00<00:00, 2784.93it/s]
100%|        | 2600/2600 [00:00<00:00, 2820.06it/s]
0%|          | 0/651 [00:00<?, ?it/s]
40%|         | 263/651 [00:00<00:00, 2626.43it/s]
100%|        | 651/651 [00:00<00:00, 2849.43it/s]
(tune_model pid=29331) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/torch/nn/modules/rnn.py:123:
UserWarning: dropout option adds dropout after all but last recurrent layer, so
non-zero dropout expects num_layers greater than 1, but got dropout=0.05 and
num_layers=1
(tune_model pid=29331) warnings.warn(
(tune_model pid=29331) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-

```



```

project/ray/issues/49454
(tune_model pid=29331) _log_deprecation_warning(
(tune_model pid=29280)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=29280) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=29361) 2025-10-01 22:10:16.975 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
(tune_model pid=29283)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=29283) warnings.warn('resource_tracker: There appear
to be %d '
    0%|          | 0/2600 [00:00<?, ?it/s]
   11%|          | 296/2600 [00:00<00:00, 2953.89it/s]
   26%|          | 680/2600 [00:00<00:00, 3473.42it/s]
   41%|          | 1054/2600 [00:00<00:00, 3578.69it/s]
   54%|          | 1412/2600 [00:00<00:00, 3080.61it/s]
   66%|          | 1729/2600 [00:00<00:00, 3059.53it/s]
   78%|          | 2041/2600 [00:00<00:00, 2902.36it/s]
  100%|          | 2600/2600 [00:00<00:00, 3171.88it/s]
    0%|          | 0/651 [00:00<?, ?it/s]
   44%|          | 286/651 [00:00<00:00, 2853.58it/s]
  100%|          | 651/651 [00:00<00:00, 3264.27it/s]
   15%|          | 382/2600 [00:00<00:00, 3819.28it/s]
   29%|          | 764/2600 [00:00<00:00, 3637.35it/s]
   50%|          | 1295/2600 [00:00<00:00, 4378.81it/s]
   67%|          | 1736/2600 [00:00<00:00, 4200.35it/s]
  100%|          | 2600/2600 [00:00<00:00, 4130.58it/s]
   63%|          | 407/651 [00:00<00:00, 4064.32it/s]
  100%|          | 651/651 [00:00<00:00, 4541.62it/s]
(tune_model pid=29331)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=29331) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=29364) 2025-10-01 22:10:18.732 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures

```

```

0%|          | 0/651 [00:00<?, ?it/s] [repeated 2x
across cluster]
(tune_model pid=29361) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=29361) _log_deprecation_warning(
(tune_model pid=29376) 2025-10-01 22:10:28.130 | INFO          |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
(tune_model pid=29281)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=29281) warnings.warn('resource_tracker: There appear
to be %d '
0%|          | 0/2600 [00:00<?, ?it/s]
19%|         | 502/2600 [00:00<00:00, 5015.80it/s]
39%|         | 1004/2600 [00:00<00:00, 4477.44it/s]
58%|         | 1512/2600 [00:00<00:00, 4732.75it/s]
76%|         | 1989/2600 [00:00<00:00, 4457.31it/s]
100%|        | 2600/2600 [00:00<00:00, 4572.17it/s]
0%|          | 0/651 [00:00<?, ?it/s]
57%|         | 370/651 [00:00<00:00, 3695.00it/s]
100%|        | 651/651 [00:00<00:00, 4248.93it/s]
(tune_model pid=29284)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=29284) warnings.warn('resource_tracker: There appear
to be %d '
18%|         | 472/2600 [00:00<00:00, 4716.70it/s]
36%|         | 944/2600 [00:00<00:00, 4404.05it/s]
54%|         | 1392/2600 [00:00<00:00, 4433.91it/s]
71%|         | 1837/2600 [00:00<00:00, 4297.55it/s]
100%|        | 2600/2600 [00:00<00:00, 4312.70it/s]
71%|         | 463/651 [00:00<00:00, 4620.48it/s]
100%|        | 651/651 [00:00<00:00, 3735.82it/s]
(tune_model pid=29364) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to

```

`ray.tune.report` when running in a function passed to Ray Tune. This will be an error in the future. See this issue for more context: <https://github.com/ray-project/ray/issues/49454>

```
(tune_model pid=29364) _log_deprecation_warning(  
(tune_model pid=29382) 2025-10-01 22:10:31.019 | INFO      |  
mads_datasets.base:download_data:121 - Folder already exists at  
/Users/nickreinders/.cache/mads_datasets/gestures  
0%|          | 0/651 [00:00<?, ?it/s] [repeated 2x
```

across cluster]

```
(tune_model pid=29390) 2025-10-01 22:10:36.478 | INFO      |  
mads_datasets.base:download_data:121 - Folder already exists at  
/Users/nickreinders/.cache/mads_datasets/gestures  
18%|         | 472/2600 [00:00<00:00, 4705.63it/s]  
36%|        | 943/2600 [00:00<00:00, 4286.51it/s]  
77%|       | 1993/2600 [00:00<00:00, 4952.27it/s]  
96%|      | 2492/2600 [00:00<00:00, 4582.95it/s]  
100%|     | 2600/2600 [00:00<00:00, 4638.70it/s]  
100%|    | 651/651 [00:00<00:00, 3601.02it/s]
```

```
(tune_model pid=29390) /Users/nickreinders/Documents/Master_Applied_Dat  
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-  
course/.venv/lib/python3.11/site-packages/torch/nn/modules/rnn.py:123:  
UserWarning: dropout option adds dropout after all but last recurrent layer, so  
non-zero dropout expects num_layers greater than 1, but got dropout=0.05 and  
num_layers=1
```

```
(tune_model pid=29390) warnings.warn(  
(tune_model pid=29376) /Users/nickreinders/Documents/Master_Applied_Dat  
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-  
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:  
RayDeprecationWarning: `ray.train.report` should be switched to  
`ray.tune.report` when running in a function passed to Ray Tune. This will be an  
error in the future. See this issue for more context: https://github.com/ray-project/ray/issues/49454
```

```
(tune_model pid=29376) _log_deprecation_warning(  
0%|          | 0/651 [00:00<?, ?it/s] [repeated 2x
```

across cluster]

```
(tune_model pid=29390) /Users/nickreinders/Documents/Master_Applied_Dat  
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-  
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:  
RayDeprecationWarning: `ray.train.report` should be switched to  
`ray.tune.report` when running in a function passed to Ray Tune. This will be an  
error in the future. See this issue for more context: https://github.com/ray-project/ray/issues/49454
```

```
(tune_model pid=29390) _log_deprecation_warning(  
(tune_model pid=29282)  
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-  
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
```

```

UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=29282) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=29418) 2025-10-01 22:10:50.021 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
 0%|          | 0/2600 [00:00<?, ?it/s]
14%|          | 352/2600 [00:00<00:00, 3510.77it/s]
27%|          | 704/2600 [00:00<00:00, 3130.22it/s]
39%|          | 1020/2600 [00:00<00:00, 3083.21it/s]
54%|          | 1399/2600 [00:00<00:00, 3347.12it/s]
69%|          | 1792/2600 [00:00<00:00, 3548.89it/s]
85%|          | 2218/2600 [00:00<00:00, 3781.53it/s]
100%|         | 2600/2600 [00:00<00:00, 3539.38it/s]
 0%|          | 0/651 [00:00<?, ?it/s]
100%|         | 651/651 [00:00<00:00, 3809.33it/s]
15%|          | 401/2600 [00:00<00:00, 4007.84it/s]
31%|          | 816/2600 [00:00<00:00, 4089.81it/s]
47%|          | 1225/2600 [00:00<00:00, 3949.33it/s]
64%|          | 1660/2600 [00:00<00:00, 4103.35it/s]
82%|          | 2145/2600 [00:00<00:00, 4367.54it/s]
100%|         | 2600/2600 [00:00<00:00, 4115.68it/s]
57%|          | 371/651 [00:00<00:00, 3696.12it/s]
100%|         | 651/651 [00:00<00:00, 3310.57it/s]
(tune_model pid=29382) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=29382) _log_deprecation_warning(
(tune_model pid=29426) 2025-10-01 22:10:53.797 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
 0%|          | 0/651 [00:00<?, ?it/s] [repeated 2x
across cluster]
(tune_model pid=29426) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=29426) _log_deprecation_warning(
(tune_model pid=29418) /Users/nickreinders/Documents/Master_Applied_Dat

```

```

a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=29418) _log_deprecation_warning(
(tune_model pid=29278)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=29278) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=29390)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=29390) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=29286)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=29286) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=29376)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown [repeated 2x across cluster]
(tune_model pid=29376) warnings.warn('resource_tracker: There appear
to be %d ' [repeated 2x across cluster]
(tune_model pid=29426)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=29426) warnings.warn('resource_tracker: There appear
to be %d '
2025-10-01 22:13:31,630 INFO tune.py:1009 -- Wrote the latest version of all
result files and experiment state to '/Users/nickreinders/Documents/Master_Appli
ed_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/notebooks/4_tuning_networks/logs/test/tune_model_2025-10-01_22-09-29' in
0.0059s.
2025-10-01 22:13:31,633 INFO tune.py:1041 -- Total run time: 242.31 seconds

```

(242.27 seconds for the tuning loop).

(tune_model pid=29382)

/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:

UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to clean up at shutdown

(tune_model pid=29382) warnings.warn('resource_tracker: There appear to be %d '

(tune_model pid=36801) 2025-10-03 13:37:59.953 | INFO |

mads_datasets.base:download_data:121 - Folder already exists at

/Users/nickreinders/.cache/mads_datasets/gestures

0%	0/2600 [00:00<?, ?it/s]
21%	535/2600 [00:00<00:00, 5340.44it/s]
41%	1070/2600 [00:00<00:00, 4850.45it/s]
63%	1639/2600 [00:00<00:00, 5211.30it/s]
100%	2600/2600 [00:00<00:00, 5343.50it/s]
0%	0/651 [00:00<?, ?it/s]
100%	651/651 [00:00<00:00, 5637.88it/s]
46%	1185/2600 [00:00<00:00, 11841.74it/s]
100%	2600/2600 [00:00<00:00, 12223.11it/s]
100%	651/651 [00:00<00:00, 12430.31it/s]
42%	1084/2600 [00:00<00:00, 10833.58it/s]
87%	2255/2600 [00:00<00:00, 11342.46it/s]
100%	2600/2600 [00:00<00:00, 11185.52it/s]
25%	639/2600 [00:00<00:00, 6387.83it/s]
58%	1519/2600 [00:00<00:00, 7803.84it/s]
100%	2600/2600 [00:00<00:00, 8056.49it/s]
100%	651/651 [00:00<00:00, 4704.09it/s]
19%	498/2600 [00:00<00:00, 4977.55it/s]
51%	1332/2600 [00:00<00:00, 6951.84it/s]
81%	2112/2600 [00:00<00:00, 7337.65it/s]
100%	2600/2600 [00:00<00:00, 6511.18it/s]
100%	651/651 [00:00<00:00, 8613.40it/s]
29%	762/2600 [00:00<00:00, 7616.43it/s]
59%	1525/2600 [00:00<00:00, 7616.48it/s]
100%	2600/2600 [00:00<00:00, 7569.03it/s]
100%	651/651 [00:00<00:00, 4925.10it/s]
17%	438/2600 [00:00<00:00, 4378.65it/s]
34%	876/2600 [00:00<00:00, 4261.10it/s]
56%	1469/2600 [00:00<00:00, 5004.57it/s]

(tune_model pid=36797) /Users/nickreinders/Documents/Master_Applied_Dat

a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-

course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:

RayDeprecationWarning: `ray.train.report` should be switched to

`ray.tune.report` when running in a function passed to Ray Tune. This will be an error in the future. See this issue for more context: <https://github.com/ray-project/ray/issues/49454>

```

(tune_model pid=36797) _log_deprecation_warning(
 76%|      | 1971/2600 [00:00<00:00, 4775.92it/s]
100%|      | 2600/2600 [00:00<00:00, 4768.29it/s]
100%|      | 651/651 [00:00<00:00, 5396.18it/s]
 17%|      | 448/2600 [00:00<00:00, 4472.04it/s]
 34%|      | 896/2600 [00:00<00:00, 3811.34it/s]
 49%|      | 1284/2600 [00:00<00:00, 3642.66it/s]
 76%|      | 1965/2600 [00:00<00:00, 4782.77it/s]
100%|      | 2600/2600 [00:00<00:00, 4749.41it/s]
100%|      | 651/651 [00:00<00:00, 7274.06it/s]
 28%|      | 720/2600 [00:00<00:00, 7194.47it/s]
(tune_model pid=36799) 2025-10-03 13:38:04.808 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 8x across
cluster]
 61%|      | 1574/2600 [00:00<00:00, 7976.05it/s]
  0%|      |  0/2600 [00:00<?, ?it/s] [repeated 15x
across cluster]
 91%|      | 2372/2600 [00:00<00:00, 7481.20it/s]
100%|      | 2600/2600 [00:00<00:00, 7345.67it/s]
 25%|      | 650/2600 [00:00<00:00, 6465.48it/s]
 81%|      | 2096/2600 [00:00<00:00, 7010.53it/s]
100%|      | 2600/2600 [00:00<00:00, 6492.71it/s]
100%|      | 651/651 [00:00<00:00, 7886.99it/s]
(tune_model pid=36801) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 9x across cluster]
(tune_model pid=36801) _log_deprecation_warning( [repeated 9x
across cluster]
(tune_model pid=36794)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=36794) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=36800) 2025-10-03 13:38:05.423 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
  0%|      |  0/651 [00:00<?, ?it/s] [repeated 3x
across cluster]

```



```

(tune_model pid=36842) 2025-10-03 13:38:15.695 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
28%|          | 721/2600 [00:00<00:00, 3554.65it/s]
42%|          | 1099/2600 [00:00<00:00, 3643.33it/s]
56%|          | 1464/2600 [00:00<00:00, 3573.50it/s]
72%|          | 1862/2600 [00:00<00:00, 3715.58it/s]
86%|          | 2245/2600 [00:00<00:00, 3753.42it/s]
100%|         | 2600/2600 [00:00<00:00, 3658.02it/s]
62%|          | 404/651 [00:00<00:00, 4035.96it/s]
100%|         | 651/651 [00:00<00:00, 3700.76it/s]
(tune_model pid=36803)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown [repeated 2x across cluster]
(tune_model pid=36803) warnings.warn('resource_tracker: There appear
to be %d ' [repeated 2x across cluster]
0%|          | 0/651 [00:00<?, ?it/s] [repeated 2x
across cluster]
19%|          | 498/2600 [00:00<00:00, 4978.00it/s]
40%|          | 1042/2600 [00:00<00:00, 5246.94it/s]
60%|          | 1567/2600 [00:00<00:00, 4610.00it/s]
79%|          | 2059/2600 [00:00<00:00, 4722.15it/s]
100%|         | 2600/2600 [00:00<00:00, 5042.13it/s]
100%|         | 651/651 [00:00<00:00, 5649.32it/s]
22%|          | 575/2600 [00:00<00:00, 5743.45it/s]
44%|          | 1150/2600 [00:00<00:00, 5130.16it/s]
68%|          | 1771/2600 [00:00<00:00, 5591.62it/s]
90%|          | 2336/2600 [00:00<00:00, 4205.31it/s]
100%|         | 2600/2600 [00:00<00:00, 4569.15it/s]
100%|         | 651/651 [00:00<00:00, 4334.86it/s]
(tune_model pid=36859) 2025-10-03 13:38:19.859 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 2x across
cluster]
18%|          | 472/2600 [00:00<00:00, 4709.58it/s]
40%|          | 1030/2600 [00:00<00:00, 5220.55it/s]
0%|          | 0/2600 [00:00<?, ?it/s] [repeated 5x
across cluster]
60%|          | 1553/2600 [00:00<00:00, 4931.37it/s]
79%|          | 2048/2600 [00:00<00:00, 4331.28it/s]
100%|         | 2600/2600 [00:00<00:00, 4638.76it/s]
100%|         | 651/651 [00:00<00:00, 4625.58it/s]
(tune_model pid=36799)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-

```



```

aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=36799) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=36859) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=36859) _log_deprecation_warning(
(tune_model pid=36872) 2025-10-03 13:38:21.943 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
(tune_model pid=36916) 2025-10-03 13:38:29.068 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
0%|          | 0/651 [00:00<?, ?it/s]
(tune_model pid=36800)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=36800) warnings.warn('resource_tracker: There appear
to be %d '
0%|          | 0/2600 [00:00<?, ?it/s]
16%|         | 409/2600 [00:00<00:00, 3953.14it/s]
36%|         | 924/2600 [00:00<00:00, 4639.27it/s]
58%|         | 1507/2600 [00:00<00:00, 5176.61it/s]
78%|         | 2026/2600 [00:00<00:00, 4839.38it/s]
(tune_model pid=36872) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 2x across cluster]
(tune_model pid=36872) _log_deprecation_warning( [repeated 2x
across cluster]
97%|         | 2514/2600 [00:00<00:00, 4199.18it/s]
100%|        | 2600/2600 [00:00<00:00, 4422.87it/s]
0%|          | 0/651 [00:00<?, ?it/s]
65%|         | 425/651 [00:00<00:00, 4248.82it/s]
100%|        | 651/651 [00:00<00:00, 4250.85it/s]
19%|         | 486/2600 [00:00<00:00, 4836.31it/s]

```

```

40%|          | 1053/2600 [00:00<00:00, 5315.09it/s]
61%|          | 1585/2600 [00:00<00:00, 4974.01it/s]
80%|          | 2085/2600 [00:00<00:00, 4860.40it/s]
100%|         | 2600/2600 [00:00<00:00, 4703.19it/s]
100%|         | 651/651 [00:00<00:00, 5454.91it/s]
37%|          | 964/2600 [00:00<00:00, 4853.42it/s]
56%|          | 1450/2600 [00:00<00:00, 4559.96it/s]
73%|          | 1908/2600 [00:00<00:00, 4297.94it/s]
92%|          | 2395/2600 [00:00<00:00, 4489.65it/s]
100%|         | 2600/2600 [00:00<00:00, 4468.48it/s]
100%|         | 651/651 [00:00<00:00, 4607.15it/s]
(tune_model pid=36919) 2025-10-03 13:38:33.036 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 2x across
cluster]
0%|           | 0/651 [00:00<?, ?it/s] [repeated 4x
across cluster]
(tune_model pid=36916) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=36916) _log_deprecation_warning(
(tune_model pid=36917) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=36917) _log_deprecation_warning(
(tune_model pid=36932) 2025-10-03 13:38:44.109 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
0%|           | 0/2600 [00:00<?, ?it/s]
18%|          | 469/2600 [00:00<00:00, 4680.21it/s]
36%|          | 938/2600 [00:00<00:00, 4435.70it/s]
53%|          | 1383/2600 [00:00<00:00, 4260.12it/s]
70%|          | 1810/2600 [00:00<00:00, 3476.95it/s]
86%|          | 2232/2600 [00:00<00:00, 3701.91it/s]
100%|         | 2600/2600 [00:00<00:00, 3858.22it/s]
0%|           | 0/651 [00:00<?, ?it/s]
47%|          | 307/651 [00:00<00:00, 3069.57it/s]
100%|         | 651/651 [00:00<00:00, 3329.09it/s]

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(tune_model pid=36843) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=36843) _log_deprecation_warning(
(tune_model pid=36859)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=36859) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=36919) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=36919) _log_deprecation_warning(
(tune_model pid=36916)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=36916) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=36932) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=36932) _log_deprecation_warning(
(tune_model pid=36917)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=36917) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=36919)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:

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UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown [repeated 2x across cluster]
(tune_model pid=36919) warnings.warn('resource_tracker: There appear
to be %d ' [repeated 2x across cluster]
(tune_model pid=36843)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=36843) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=38535) 2025-10-03 14:35:50.435 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
 0%|          | 0/2600 [00:00<?, ?it/s]
19%|          | 498/2600 [00:00<00:00, 4975.22it/s]
40%|          | 1040/2600 [00:00<00:00, 5228.97it/s]
60%|          | 1563/2600 [00:00<00:00, 5204.18it/s]
80%|          | 2084/2600 [00:00<00:00, 5174.92it/s]
100%|         | 2600/2600 [00:00<00:00, 5193.92it/s]
 0%|          | 0/651 [00:00<?, ?it/s]
100%|         | 651/651 [00:00<00:00, 5565.76it/s]
42%|          | 1090/2600 [00:00<00:00, 10891.21it/s]
100%|         | 2600/2600 [00:00<00:00, 11034.96it/s]
100%|         | 651/651 [00:00<00:00, 10419.02it/s]
26%|          | 683/2600 [00:00<00:00, 6829.47it/s]
59%|          | 1531/2600 [00:00<00:00, 7799.28it/s]
100%|         | 2600/2600 [00:00<00:00, 7784.03it/s]
100%|         | 651/651 [00:00<00:00, 8228.31it/s]
(tune_model pid=38540) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=38540) _log_deprecation_warning(
(tune_model pid=38542) 2025-10-03 14:35:55.258 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 3x across
cluster]
 0%|          | 0/2600 [00:00<?, ?it/s] [repeated 5x
across cluster]
18%|          | 471/2600 [00:00<00:00, 4707.03it/s]
42%|          | 1083/2600 [00:00<00:00, 5535.72it/s]
100%|         | 2600/2600 [00:00<00:00, 6991.69it/s]

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100%|      | 651/651 [00:00<00:00, 7741.35it/s]
 64%|      | 1663/2600 [00:00<00:00, 8445.56it/s]
100%|      | 2600/2600 [00:00<00:00, 8400.13it/s]
100%|      | 651/651 [00:00<00:00, 7305.98it/s]
 28%|      | 722/2600 [00:00<00:00, 7193.44it/s]
 55%|      | 1442/2600 [00:00<00:00, 6860.36it/s]
 87%|      | 2259/2600 [00:00<00:00, 7439.32it/s]
100%|      | 2600/2600 [00:00<00:00, 6685.31it/s]
100%|      | 651/651 [00:00<00:00, 6467.43it/s]
(tune_model pid=38542) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 2x across cluster]
(tune_model pid=38542) _log_deprecation_warning( [repeated 2x
across cluster]
(tune_model pid=38556) 2025-10-03 14:36:01.502 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 3x across
cluster]
 0%|      | 0/651 [00:00<?, ?it/s] [repeated 5x
across cluster]
 46%|      | 1203/2600 [00:00<00:00, 6039.69it/s]
 70%|      | 1807/2600 [00:00<00:00, 5005.93it/s]
100%|      | 2600/2600 [00:00<00:00, 5203.71it/s]
100%|      | 651/651 [00:00<00:00, 5912.11it/s]
 49%|      | 1284/2600 [00:00<00:00, 6266.52it/s]
 74%|      | 1915/2600 [00:00<00:00, 6285.92it/s]
100%|      | 2600/2600 [00:00<00:00, 5497.32it/s]
100%|      | 651/651 [00:00<00:00, 6350.66it/s]
(tune_model pid=38540)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=38540) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=38538) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 2x across cluster]

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(tune_model pid=38538) _log_deprecation_warning( [repeated 2x
across cluster]
(tune_model pid=38576) 2025-10-03 14:36:06.775 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 2x across
cluster]
  0%|          | 0/651 [00:00<?, ?it/s] [repeated 4x
across cluster]
 21%|          | 536/2600 [00:00<00:00, 5354.27it/s]
 41%|          | 1072/2600 [00:00<00:00, 5175.40it/s]
 61%|          | 1590/2600 [00:00<00:00, 5152.68it/s]
 81%|          | 2115/2600 [00:00<00:00, 5188.38it/s]
100%|          | 2600/2600 [00:00<00:00, 5124.20it/s]
100%|          | 651/651 [00:00<00:00, 6348.86it/s]
 15%|          | 391/2600 [00:00<00:00, 3760.54it/s]
 34%|          | 884/2600 [00:00<00:00, 4435.26it/s]
 56%|          | 1466/2600 [00:00<00:00, 5058.81it/s]
 77%|          | 1998/2600 [00:00<00:00, 5161.07it/s]
100%|          | 2600/2600 [00:00<00:00, 4917.99it/s]
100%|          | 651/651 [00:00<00:00, 5925.83it/s]
(tune_model pid=38576) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 4x across cluster]
(tune_model pid=38576) _log_deprecation_warning( [repeated 4x
across cluster]
(tune_model pid=38582) 2025-10-03 14:36:09.215 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
  0%|          | 0/651 [00:00<?, ?it/s] [repeated 4x
across cluster]
(tune_model pid=38572)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=38572) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=38582) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to

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`ray.tune.report` when running in a function passed to Ray Tune. This will be an error in the future. See this issue for more context: <https://github.com/ray-project/ray/issues/49454>

```
(tune_model pid=38582) _log_deprecation_warning(
(tune_model pid=38550)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=38550) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=38576)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=38576) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=38556)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown [repeated 2x across cluster]
(tune_model pid=38556) warnings.warn('resource_tracker: There appear
to be %d ' [repeated 2x across cluster]
(tune_model pid=38535)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=38535) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=38548)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=38548) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=38538)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=38538) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=38840) 2025-10-03 14:37:46.281 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
```



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/Users/nickreinders/.cache/mads_datasets/gestures
 0%|          | 0/2600 [00:00<?, ?it/s]
19%|          | 504/2600 [00:00<00:00, 5034.40it/s]
40%|          | 1029/2600 [00:00<00:00, 5155.50it/s]
61%|          | 1593/2600 [00:00<00:00, 5374.84it/s]
82%|          | 2131/2600 [00:00<00:00, 5286.50it/s]
100%|         | 2600/2600 [00:00<00:00, 5255.39it/s]
 0%|          | 0/651 [00:00<?, ?it/s]
100%|         | 651/651 [00:00<00:00, 4507.00it/s]
40%|          | 1040/2600 [00:00<00:00, 10395.95it/s]
80%|          | 2086/2600 [00:00<00:00, 10433.14it/s]
100%|         | 2600/2600 [00:00<00:00, 10403.04it/s]
(tune_model pid=38841) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/torch/nn/modules/rnn.py:123:
UserWarning: dropout option adds dropout after all but last recurrent layer, so
non-zero dropout expects num_layers greater than 1, but got dropout=0.1 and
num_layers=1
(tune_model pid=38841) warnings.warn(
(tune_model pid=38841) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=38841) _log_deprecation_warning(
32%|          | 842/2600 [00:00<00:00, 8412.85it/s]
65%|          | 1684/2600 [00:00<00:00, 7877.67it/s]
100%|         | 2600/2600 [00:00<00:00, 8182.20it/s]
(tune_model pid=38845) 2025-10-03 14:37:51.343 | INFO          |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 3x across
cluster]
100%|         | 651/651 [00:00<00:00, 8309.82it/s]

[repeated 4x across cluster]
27%|          | 712/2600 [00:00<00:00, 7111.77it/s]
55%|          | 1424/2600 [00:00<00:00, 7036.11it/s]
100%|         | 2600/2600 [00:00<00:00, 7310.60it/s]
100%|         | 651/651 [00:00<00:00, 5891.31it/s]
19%|          | 495/2600 [00:00<00:00, 4945.10it/s]
46%|          | 1193/2600 [00:00<00:00, 6140.53it/s]
70%|          | 1832/2600 [00:00<00:00, 6249.30it/s]
100%|         | 2600/2600 [00:00<00:00, 6145.90it/s]
100%|         | 651/651 [00:00<00:00, 6032.32it/s]
(tune_model pid=38843) /Users/nickreinders/Documents/Master_Applied_Dat

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a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 2x across cluster]
(tune_model pid=38843) _log_deprecation_warning( [repeated 2x
across cluster]
24%|          | 623/2600 [00:00<00:00, 6225.39it/s]
48%|          | 1246/2600 [00:00<00:00, 6193.79it/s]
72%|          | 1866/2600 [00:00<00:00, 5973.39it/s]
100%|         | 2600/2600 [00:00<00:00, 5869.79it/s]
100%|         | 651/651 [00:00<00:00, 6181.79it/s]
(tune_model pid=38867) 2025-10-03 14:37:55.769 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 2x across
cluster]
0%|          | 0/651 [00:00<?, ?it/s] [repeated 6x
across cluster]
20%|          | 513/2600 [00:00<00:00, 5127.85it/s]
39%|          | 1026/2600 [00:00<00:00, 4311.24it/s]
58%|          | 1504/2600 [00:00<00:00, 4504.21it/s]
75%|          | 1961/2600 [00:00<00:00, 3832.66it/s]
93%|          | 2419/2600 [00:00<00:00, 4061.37it/s]
(tune_model pid=38845) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=38845) _log_deprecation_warning(
100%|         | 2600/2600 [00:00<00:00, 4063.32it/s]
56%|          | 365/651 [00:00<00:00, 3641.15it/s]
100%|         | 651/651 [00:00<00:00, 3296.58it/s]
(tune_model pid=38865) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=38865) _log_deprecation_warning(
(tune_model pid=38875) 2025-10-03 14:38:01.922 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at

```

```

/Users/nickreinders/.cache/mads_datasets/gestures [repeated 2x across
cluster]
  0%|          | 0/651 [00:00<?, ?it/s] [repeated 2x
across cluster]
 19%|          | 501/2600 [00:00<00:00, 5003.80it/s]
 39%|          | 1018/2600 [00:00<00:00, 5096.87it/s]
 59%|          | 1528/2600 [00:00<00:00, 4854.48it/s]
 78%|          | 2015/2600 [00:00<00:00, 4567.77it/s]
100%|          | 2600/2600 [00:00<00:00, 4880.17it/s]
100%|          | 651/651 [00:00<00:00, 5201.17it/s]
(tune_model pid=38873) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 2x across cluster]
(tune_model pid=38873) _log_deprecation_warning( [repeated 2x
across cluster]
  0%|          | 0/651 [00:00<?, ?it/s] [repeated 2x
across cluster]
(tune_model pid=38840)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=38840) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=38875) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=38875) _log_deprecation_warning(
(tune_model pid=38845)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown [repeated 3x across cluster]
(tune_model pid=38845) warnings.warn('resource_tracker: There appear
to be %d ' [repeated 3x across cluster]
(tune_model pid=38875)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-

```

```

aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=38875) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=41714) 2025-10-03 20:07:39.892 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
 0%|          | 0/2600 [00:00<?, ?it/s]
15%|          | 395/2600 [00:00<00:00, 3940.94it/s]
35%|          | 909/2600 [00:00<00:00, 4640.44it/s]
56%|          | 1467/2600 [00:00<00:00, 5068.62it/s]
76%|          | 1974/2600 [00:00<00:00, 4959.30it/s]
96%|          | 2484/2600 [00:00<00:00, 5008.83it/s]
100%|         | 2600/2600 [00:00<00:00, 4938.11it/s]
 84%|          | 548/651 [00:00<00:00, 5473.70it/s]
100%|         | 651/651 [00:00<00:00, 5438.05it/s]
 47%|          | 1234/2600 [00:00<00:00, 12334.45it/s]
100%|         | 2600/2600 [00:00<00:00, 12337.36it/s]
100%|         | 651/651 [00:00<00:00, 8387.24it/s]
 41%|          | 1074/2600 [00:00<00:00, 10721.89it/s]
100%|         | 2600/2600 [00:00<00:00, 11275.33it/s]
 35%|          | 910/2600 [00:00<00:00, 9092.94it/s]
 75%|          | 1939/2600 [00:00<00:00, 9795.49it/s]
100%|         | 2600/2600 [00:00<00:00, 9716.84it/s]
 30%|          | 784/2600 [00:00<00:00, 7828.88it/s]
 60%|          | 1567/2600 [00:00<00:00, 7447.36it/s]
100%|         | 2600/2600 [00:00<00:00, 8017.54it/s]
(tune_model pid=41720) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/torch/nn/modules/rnn.py:123:
UserWarning: dropout option adds dropout after all but last recurrent layer, so
non-zero dropout expects num_layers greater than 1, but got dropout=0.1 and
num_layers=1
(tune_model pid=41720) warnings.warn(
 31%|          | 814/2600 [00:00<00:00, 8134.05it/s]
 65%|          | 1699/2600 [00:00<00:00, 8549.89it/s]
100%|         | 2600/2600 [00:00<00:00, 8562.06it/s]
(tune_model pid=41714) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=41714) _log_deprecation_warning(
 52%|          | 1356/2600 [00:00<00:00, 7034.36it/s]
 81%|          | 2111/2600 [00:00<00:00, 7269.32it/s]

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100%|      | 2600/2600 [00:00<00:00, 7287.88it/s]
(tune_model pid=41720)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=41720) warnings.warn('resource_tracker: There appear
to be %d '
100%|      | 651/651 [00:00<00:00, 6566.73it/s]
26%|      | 688/2600 [00:00<00:00, 6872.71it/s]
59%|      | 1532/2600 [00:00<00:00, 7783.49it/s]
89%|      | 2311/2600 [00:00<00:00, 7727.66it/s]
100%|      | 2600/2600 [00:00<00:00, 7774.12it/s]
17%|      | 433/2600 [00:00<00:00, 4300.71it/s]
36%|      | 940/2600 [00:00<00:00, 4743.45it/s]
59%|      | 1532/2600 [00:00<00:00, 5274.88it/s]
79%|      | 2060/2600 [00:00<00:00, 4750.31it/s]
100%|      | 2600/2600 [00:00<00:00, 5152.70it/s]
100%|      | 651/651 [00:00<00:00, 6906.87it/s]
(tune_model pid=41719) 2025-10-03 20:07:44.849 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 9x across
cluster]
15%|      | 388/2600 [00:00<00:00, 3876.67it/s]
0%|      | 0/2600 [00:00<?, ?it/s] [repeated 17x]

across cluster]
40%|      | 1052/2600 [00:00<00:00, 5492.28it/s]
65%|      | 1689/2600 [00:00<00:00, 5889.10it/s]
88%|      | 2278/2600 [00:00<00:00, 5452.38it/s]
100%|      | 2600/2600 [00:00<00:00, 4583.58it/s]
37%|      | 244/651 [00:00<00:00, 2311.88it/s]
100%|      | 651/651 [00:00<00:00, 3484.27it/s]
(tune_model pid=41721) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 3x across cluster]
(tune_model pid=41721) _log_deprecation_warning( [repeated 3x]

across cluster]
(tune_model pid=41721)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown

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(tune_model pid=41721) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=41739) 2025-10-03 20:07:49.784 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
18%|          | 455/2600 [00:00<00:00, 4547.36it/s]
35%|          | 910/2600 [00:00<00:00, 4314.77it/s]
0%|          | 0/2600 [00:00<?, ?it/s] [repeated 2x
across cluster]
56%|          | 1467/2600 [00:00<00:00, 4859.72it/s]
75%|          | 1955/2600 [00:00<00:00, 4447.24it/s]
100%|         | 2600/2600 [00:00<00:00, 4517.18it/s]
100%|         | 651/651 [00:00<00:00, 5488.29it/s]
(tune_model pid=41758) 2025-10-03 20:07:53.580 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
21%|          | 538/2600 [00:00<00:00, 5379.24it/s]
41%|          | 1076/2600 [00:00<00:00, 4736.54it/s]
60%|          | 1555/2600 [00:00<00:00, 4621.37it/s]
78%|          | 2020/2600 [00:00<00:00, 4322.99it/s]
100%|         | 2600/2600 [00:00<00:00, 4840.00it/s]
100%|         | 651/651 [00:00<00:00, 5301.30it/s]
(tune_model pid=41718) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 2x across cluster]
(tune_model pid=41718) _log_deprecation_warning( [repeated 2x
across cluster]
0%|          | 0/651 [00:00<?, ?it/s] [repeated 3x
across cluster]
(tune_model pid=41717) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 2x across cluster]
(tune_model pid=41717) _log_deprecation_warning( [repeated 2x
across cluster]
(tune_model pid=41714)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:

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UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=41714) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=41739) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 2x across cluster]
(tune_model pid=41739) _log_deprecation_warning( [repeated 2x
across cluster]
(tune_model pid=41783) 2025-10-03 20:08:17.111 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
0%|          | 0/2600 [00:00<?, ?it/s]
9%|          | 225/2600 [00:00<00:01, 2249.42it/s]
18%|         | 475/2600 [00:00<00:00, 2396.42it/s]
32%|         | 820/2600 [00:00<00:00, 2874.95it/s]
45%|         | 1177/2600 [00:00<00:00, 3145.56it/s]
60%|         | 1557/2600 [00:00<00:00, 3378.29it/s]
73%|         | 1895/2600 [00:00<00:00, 3181.83it/s]
86%|         | 2223/2600 [00:00<00:00, 3210.71it/s]
100%|        | 2600/2600 [00:00<00:00, 3132.14it/s]
0%|          | 0/651 [00:00<?, ?it/s]
49%|         | 318/651 [00:00<00:00, 3174.51it/s]
100%|        | 651/651 [00:00<00:00, 2794.26it/s]
(tune_model pid=41712) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=41712) _log_deprecation_warning(
(tune_model pid=41718)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=41718) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=41823) 2025-10-03 20:08:36.903 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
(tune_model pid=41719)

```

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/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=41719) warnings.warn('resource_tracker: There appear
to be %d '
  0%|          | 0/2600 [00:00<?, ?it/s]
 10%|          | 258/2600 [00:00<00:00, 2579.26it/s]
 20%|          | 522/2600 [00:00<00:00, 2612.53it/s]
 30%|          | 784/2600 [00:00<00:00, 2071.66it/s]
 44%|          | 1136/2600 [00:00<00:00, 2562.74it/s]
 54%|          | 1407/2600 [00:00<00:00, 2490.14it/s]
 68%|          | 1771/2600 [00:00<00:00, 2844.54it/s]
 79%|          | 2066/2600 [00:00<00:00, 2602.61it/s]
100%|         | 2600/2600 [00:00<00:00, 2621.80it/s]
  0%|          | 0/651 [00:00<?, ?it/s]
 49%|          | 316/651 [00:00<00:00, 3152.74it/s]
100%|         | 651/651 [00:00<00:00, 3277.12it/s]
 16%|          | 404/2600 [00:00<00:00, 4033.68it/s]
 31%|          | 808/2600 [00:00<00:00, 2604.45it/s]
 45%|          | 1173/2600 [00:00<00:00, 2404.96it/s]
 55%|          | 1430/2600 [00:00<00:00, 2350.38it/s]
 70%|          | 1817/2600 [00:00<00:00, 2780.55it/s]
 89%|          | 2305/2600 [00:00<00:00, 3322.75it/s]
100%|         | 2600/2600 [00:00<00:00, 2988.51it/s]
 60%|          | 390/651 [00:00<00:00, 3846.42it/s]
100%|         | 651/651 [00:00<00:00, 3385.21it/s]
(tune_model pid=41783) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=41783) _log_deprecation_warning(
(tune_model pid=41783)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=41783) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=41822) 2025-10-03 20:08:38.375 | INFO          |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
  0%|          | 0/651 [00:00<?, ?it/s] [repeated 2x
across cluster]

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(tune_model pid=41857) 2025-10-03 20:08:47.446 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
(tune_model pid=41715) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 2x across cluster]
(tune_model pid=41715) _log_deprecation_warning( [repeated 2x
across cluster]
15%|          | 401/2600 [00:00<00:00, 4002.71it/s]
31%|          | 802/2600 [00:00<00:00, 3473.59it/s]
46%|          | 1203/2600 [00:00<00:00, 3693.64it/s]
62%|          | 1599/2600 [00:00<00:00, 3788.62it/s]
76%|          | 1983/2600 [00:00<00:00, 3776.59it/s]
100%|         | 2600/2600 [00:00<00:00, 3889.26it/s]
70%|          | 455/651 [00:00<00:00, 4541.87it/s]
100%|         | 651/651 [00:00<00:00, 4198.08it/s]
0%|           | 0/651 [00:00<?, ?it/s] [repeated 2x
across cluster]
12%|          | 311/2600 [00:00<00:00, 3107.05it/s]
24%|          | 622/2600 [00:00<00:00, 2998.06it/s]
56%|          | 1445/2600 [00:00<00:00, 3717.54it/s]
87%|          | 2273/2600 [00:00<00:00, 3839.96it/s]
100%|         | 2600/2600 [00:00<00:00, 3611.67it/s]
100%|         | 651/651 [00:00<00:00, 4601.09it/s]
(tune_model pid=41867) 2025-10-03 20:08:54.041 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 2x across
cluster]
(tune_model pid=41822) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=41822) _log_deprecation_warning(
13%|          | 340/2600 [00:00<00:00, 3358.13it/s]
33%|          | 854/2600 [00:00<00:00, 4388.81it/s]
50%|          | 1294/2600 [00:00<00:00, 4200.49it/s]
66%|          | 1716/2600 [00:00<00:00, 3856.27it/s]
85%|          | 2217/2600 [00:00<00:00, 4237.04it/s]
100%|         | 2600/2600 [00:00<00:00, 4055.54it/s]

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60%|          | 389/651 [00:00<00:00, 3887.55it/s]
100%|         | 651/651 [00:00<00:00, 3804.81it/s]
(tune_model pid=41867) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/torch/nn/modules/rnn.py:123:
UserWarning: dropout option adds dropout after all but last recurrent layer, so non-zero dropout expects num_layers greater than 1, but got dropout=0.1 and num_layers=1
(tune_model pid=41867)   warnings.warn(
(tune_model pid=41716)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to clean up at shutdown
(tune_model pid=41716)   warnings.warn('resource_tracker: There appear to be %d '
0%|          | 0/651 [00:00<?, ?it/s] [repeated 4x
across cluster]
(tune_model pid=41857) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to `ray.tune.report` when running in a function passed to Ray Tune. This will be an error in the future. See this issue for more context: https://github.com/ray-project/ray/issues/49454
(tune_model pid=41857)   _log_deprecation_warning(
(tune_model pid=41713)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to clean up at shutdown [repeated 3x across cluster]
(tune_model pid=41713)   warnings.warn('resource_tracker: There appear to be %d ' [repeated 3x across cluster]
(tune_model pid=41859) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to `ray.tune.report` when running in a function passed to Ray Tune. This will be an error in the future. See this issue for more context: https://github.com/ray-project/ray/issues/49454 [repeated 2x across cluster]
(tune_model pid=41859)   _log_deprecation_warning( [repeated 2x
across cluster]
(tune_model pid=41857)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to

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clean up at shutdown [repeated 2x across cluster]
(tune_model pid=41857) warnings.warn('resource_tracker: There appear
to be %d ' [repeated 2x across cluster]
(tune_model pid=41715)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown [repeated 2x across cluster]
(tune_model pid=41715) warnings.warn('resource_tracker: There appear
to be %d ' [repeated 2x across cluster]
(tune_model pid=41717)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=41717) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=50239) 2025-10-04 08:20:25.263 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
 0%|          | 0/2600 [00:00<?, ?it/s]
19%|          | 495/2600 [00:00<00:00, 4948.27it/s]
40%|          | 1030/2600 [00:00<00:00, 5180.03it/s]
61%|          | 1586/2600 [00:00<00:00, 5351.65it/s]
82%|          | 2129/2600 [00:00<00:00, 5379.85it/s]
100%|         | 2600/2600 [00:00<00:00, 5297.92it/s]
 0%|          | 0/651 [00:00<?, ?it/s]
100%|         | 651/651 [00:00<00:00, 5298.42it/s]
 82%|          | 2131/2600 [00:00<00:00, 10836.09it/s]
100%|         | 2600/2600 [00:00<00:00, 10756.11it/s]
 29%|          | 750/2600 [00:00<00:00, 7492.13it/s]
 66%|          | 1719/2600 [00:00<00:00, 8782.67it/s]
100%|         | 2600/2600 [00:00<00:00, 8999.65it/s]
100%|         | 651/651 [00:00<00:00, 5985.94it/s]
(tune_model pid=50246) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/torch/nn/modules/rnn.py:123:
UserWarning: dropout option adds dropout after all but last recurrent layer, so
non-zero dropout expects num_layers greater than 1, but got dropout=0.1 and
num_layers=1
(tune_model pid=50246) warnings.warn(
(tune_model pid=50244) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454

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(tune_model pid=50244) _log_deprecation_warning(
  66%|          | 1728/2600 [00:00<00:00, 8748.23it/s]
100%|          | 2600/2600 [00:00<00:00, 8833.62it/s]
(tune_model pid=50246)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=50246) warnings.warn('resource_tracker: There appear
to be %d '
100%|          | 651/651 [00:00<00:00, 9358.15it/s]
(tune_model pid=50250) 2025-10-04 08:20:31.485 | INFO          |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 4x across
cluster]
  0%|          | 0/651 [00:00<?, ?it/s] [repeated 6x
across cluster]
 24%|          | 618/2600 [00:00<00:00, 6179.25it/s]
 54%|          | 1407/2600 [00:00<00:00, 7182.40it/s]
100%|          | 2600/2600 [00:00<00:00, 7559.02it/s]
100%|          | 651/651 [00:00<00:00, 8261.10it/s]
 18%|          | 474/2600 [00:00<00:00, 4736.55it/s]
 42%|          | 1105/2600 [00:00<00:00, 5660.07it/s]
 66%|          | 1713/2600 [00:00<00:00, 5846.92it/s]
(tune_model pid=50246) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=50246) _log_deprecation_warning(
100%|          | 2600/2600 [00:00<00:00, 5611.08it/s]
100%|          | 651/651 [00:00<00:00, 4931.45it/s]
(tune_model pid=50250) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=50250) _log_deprecation_warning(
(tune_model pid=50250)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to

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clean up at shutdown
(tune_model pid=50250) warnings.warn('resource_tracker: There appear
to be %d '
27%|          | 700/2600 [00:00<00:00, 6995.09it/s]
56%|          | 1455/2600 [00:00<00:00, 7321.27it/s]
100%|         | 2600/2600 [00:00<00:00, 7133.51it/s]
100%|         | 651/651 [00:00<00:00, 7149.70it/s]
(tune_model pid=50258) 2025-10-04 08:20:35.978 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 2x across
cluster]
0%|           | 0/651 [00:00<?, ?it/s] [repeated 6x
across cluster]
23%|          | 589/2600 [00:00<00:00, 5886.76it/s]
49%|          | 1280/2600 [00:00<00:00, 6487.69it/s]
100%|         | 2600/2600 [00:00<00:00, 6469.05it/s]
100%|         | 651/651 [00:00<00:00, 6015.97it/s]
(tune_model pid=50280) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=50280) _log_deprecation_warning(
18%|          | 468/2600 [00:00<00:00, 4678.63it/s]
(tune_model pid=50280)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=50280) warnings.warn('resource_tracker: There appear
to be %d '
45%|          | 1160/2600 [00:00<00:00, 5995.66it/s]
70%|          | 1812/2600 [00:00<00:00, 6233.73it/s]
100%|         | 2600/2600 [00:00<00:00, 5850.96it/s]
100%|         | 651/651 [00:00<00:00, 6174.82it/s]
(tune_model pid=50286) 2025-10-04 08:20:43.269 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 3x across
cluster]
0%|           | 0/651 [00:00<?, ?it/s] [repeated 4x
across cluster]
10%|          | 272/2600 [00:00<00:00, 2719.21it/s]
30%|          | 782/2600 [00:00<00:00, 4118.99it/s]

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76%|          | 1968/2600 [00:00<00:00, 5305.75it/s]
100%|         | 2600/2600 [00:00<00:00, 4903.47it/s]
100%|         | 651/651 [00:00<00:00, 5495.99it/s]
(tune_model pid=50248) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 2x across cluster]
(tune_model pid=50248) _log_deprecation_warning( [repeated 2x
across cluster]
22%|          | 559/2600 [00:00<00:00, 5586.37it/s]
43%|          | 1118/2600 [00:00<00:00, 4683.30it/s]
67%|          | 1731/2600 [00:00<00:00, 5279.70it/s]
100%|         | 2600/2600 [00:00<00:00, 5110.46it/s]
100%|         | 651/651 [00:00<00:00, 4518.36it/s]
(tune_model pid=50297) 2025-10-04 08:20:49.189 | INFO          |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures [repeated 2x across
cluster]
0%|            | 0/651 [00:00<?, ?it/s] [repeated 4x
across cluster]
23%|          | 604/2600 [00:00<00:00, 6039.03it/s]
(tune_model pid=50286)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=50286) warnings.warn('resource_tracker: There appear
to be %d '
46%|          | 1208/2600 [00:00<00:00, 4742.65it/s]
70%|          | 1815/2600 [00:00<00:00, 5262.80it/s]
91%|          | 2359/2600 [00:00<00:00, 4850.31it/s]
100%|         | 2600/2600 [00:00<00:00, 4930.36it/s]
100%|         | 651/651 [00:00<00:00, 5008.54it/s]
(tune_model pid=50286) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=50286) _log_deprecation_warning(
40%|          | 1040/2600 [00:00<00:00, 5240.07it/s]
60%|          | 1564/2600 [00:00<00:00, 4690.71it/s]

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78%|      | 2040/2600 [00:00<00:00, 3924.80it/s]
98%|      | 2551/2600 [00:00<00:00, 4284.61it/s]
100%|     | 2600/2600 [00:00<00:00, 4433.98it/s]
70%|      | 456/651 [00:00<00:00, 4544.69it/s]
100%|     | 651/651 [00:00<00:00, 4278.23it/s]
(tune_model pid=50292) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=50292) _log_deprecation_warning(
(tune_model pid=50316) 2025-10-04 08:20:51.994 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
0%|      | 0/651 [00:00<?, ?it/s] [repeated 4x
across cluster]
(tune_model pid=50323) 2025-10-04 08:20:55.516 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
(tune_model pid=50292)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=50292) warnings.warn('resource_tracker: There appear
to be %d '
18%|      | 479/2600 [00:00<00:00, 4338.21it/s]
36%|      | 935/2600 [00:00<00:00, 4462.81it/s]
53%|      | 1382/2600 [00:00<00:00, 4343.54it/s]
70%|      | 1817/2600 [00:00<00:00, 4076.53it/s]
87%|      | 2252/2600 [00:00<00:00, 4018.86it/s]
100%|     | 2600/2600 [00:00<00:00, 4249.27it/s]
100%|     | 651/651 [00:00<00:00, 6008.98it/s]
(tune_model pid=50316) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=50316) _log_deprecation_warning(
0%|      | 0/651 [00:00<?, ?it/s] [repeated 2x
across cluster]
11%|      | 293/2600 [00:00<00:00, 2928.33it/s]

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23%|          | 586/2600 [00:00<00:00, 2154.62it/s]
37%|          | 962/2600 [00:00<00:00, 2658.04it/s]
(tune_model pid=50326) 2025-10-04 08:20:59.948 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
56%|          | 1443/2600 [00:00<00:00, 3394.23it/s]
74%|          | 1926/2600 [00:00<00:00, 3866.20it/s]
91%|          | 2368/2600 [00:00<00:00, 4042.93it/s]
100%|         | 2600/2600 [00:00<00:00, 3370.15it/s]
57%|          | 373/651 [00:00<00:00, 3729.28it/s]
100%|         | 651/651 [00:00<00:00, 2915.79it/s]
(tune_model pid=50284) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=50284) _log_deprecation_warning(
(tune_model pid=50297)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=50297) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=50335) 2025-10-04 08:21:04.245 | INFO      |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
17%|          | 448/2600 [00:00<00:00, 4441.45it/s]
37%|          | 954/2600 [00:00<00:00, 4792.57it/s]
56%|          | 1443/2600 [00:00<00:00, 4835.14it/s]
74%|          | 1927/2600 [00:00<00:00, 4440.92it/s]
100%|         | 2600/2600 [00:00<00:00, 4438.70it/s]
0%|           | 0/651 [00:00<?, ?it/s] [repeated 4x
across cluster]
100%|         | 651/651 [00:00<00:00, 5208.08it/s]
(tune_model pid=50335) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/torch/nn/modules/rnn.py:123:
UserWarning: dropout option adds dropout after all but last recurrent layer, so
non-zero dropout expects num_layers greater than 1, but got dropout=0.1 and
num_layers=1
(tune_model pid=50335) warnings.warn(
21%|          | 554/2600 [00:00<00:00, 5538.41it/s]
43%|          | 1108/2600 [00:00<00:00, 5445.48it/s]
64%|          | 1653/2600 [00:00<00:00, 4796.25it/s]

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86%|          | 2226/2600 [00:00<00:00, 5134.61it/s]
(tune_model pid=50256) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 3x across cluster]
(tune_model pid=50256) _log_deprecation_warning( [repeated 3x
across cluster]
100%|          | 2600/2600 [00:00<00:00, 5056.63it/s]
82%|          | 536/651 [00:00<00:00, 5356.63it/s]
100%|          | 651/651 [00:00<00:00, 5367.57it/s]
(tune_model pid=50326)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=50326) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=50357) 2025-10-04 08:21:07.881 | INFO          |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
0%|          | 0/651 [00:00<?, ?it/s] [repeated 2x
across cluster]
(tune_model pid=50335)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=50335) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=50363) 2025-10-04 08:21:11.459 | INFO          |
mads_datasets.base:download_data:121 - Folder already exists at
/Users/nickreinders/.cache/mads_datasets/gestures
11%|          | 285/2600 [00:00<00:00, 2848.83it/s]
23%|          | 602/2600 [00:00<00:00, 3035.44it/s]
40%|          | 1046/2600 [00:00<00:00, 3642.43it/s]
55%|          | 1417/2600 [00:00<00:00, 3665.75it/s]
69%|          | 1784/2600 [00:00<00:00, 3454.49it/s]
87%|          | 2267/2600 [00:00<00:00, 3896.98it/s]
100%|          | 2600/2600 [00:00<00:00, 3730.63it/s]
100%|          | 651/651 [00:00<00:00, 3721.81it/s]
(tune_model pid=50258) /Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:

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RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454 [repeated 2x across cluster]
(tune_model pid=50258) _log_deprecation_warning( [repeated 2x
across cluster]
0%|          | 0/651 [00:00<?, ?it/s] [repeated 2x
across cluster]
(tune_model pid=50357) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=50357) _log_deprecation_warning(
(tune_model pid=50323) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=50323) _log_deprecation_warning(
(tune_model pid=50363) /Users/nickreinders/Documents/Master_Applied_Dat
a_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/.venv/lib/python3.11/site-packages/ray/train/_internal/session.py:772:
RayDeprecationWarning: `ray.train.report` should be switched to
`ray.tune.report` when running in a function passed to Ray Tune. This will be an
error in the future. See this issue for more context: https://github.com/ray-
project/ray/issues/49454
(tune_model pid=50363) _log_deprecation_warning(
(tune_model pid=50363)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=50363) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=50357)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=50357) warnings.warn('resource_tracker: There appear
to be %d '

```

```

(tune_model pid=50248)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=50248) warnings.warn('resource_tracker: There appear
to be %d '
(tune_model pid=50258)
/Users/nickreinders/.local/share/uv/python/cpython-3.11.9-macos-
aarch64-none/lib/python3.11/multiprocessing/resource_tracker.py:254:
UserWarning: resource_tracker: There appear to be 1 leaked semaphore objects to
clean up at shutdown
(tune_model pid=50258) warnings.warn('resource_tracker: There appear
to be %d '

```

```

[13]: def plot_contour(df, x, y, z, start=0.90, end=1.0, size=0.01):
    fig = go.Figure()

    fig.add_trace(
        go.Contour(
            z=df[z],
            x=df[x],
            y=df[y],
            contours=dict(
                coloring='heatmap',
                showlabels=True, # show labels on contours
                start=start,     # start of the contour range
                end=end,         # end of the contour range
                size=size,
            ),
            colorscale="plasma",
            colorbar=dict(
                title='Accuracy'
            )
        )
    )

    fig.add_trace(
        go.Scatter(
            x=df[x],
            y=df[y],
            mode='markers',
            marker=dict(
                color='black',
                size=8,
                symbol='circle'
            ),

```

```

        customdata=df['accuracy'], # Pass accuracy values for hover text
        hovertemplate=(
            'Hidden Size: %{x}<br>'
            'Number of Layers: %{y}<br>'
            'Accuracy: %{customdata:.4f}<extra></extra>'
        ),
        name='Data Points'
    )
)

fig.update_layout(
    title="Contour Plot",
    xaxis_title="Hidden Size",
    yaxis_title="Number of Layers",
    xaxis=dict(showgrid=False), # Remove x-axis grid lines
    yaxis=dict(showgrid=False),
    plot_bgcolor='white',      # Set background color to white
    paper_bgcolor='white'
)

fig.show()

```

```
[14]: all_results = analysis.results_df
```

```
[15]: random = analysis.results_df
      plot_contour(random, "config/hidden_size", "config/num_layers", "accuracy")
```

As you can see, the search space is sort of randomly sampled. Even though big parts are unexplored, it still looks like we find some hotspots.

```
[16]: best = analysis.get_best_config()
      best["accuracy"] = analysis.best_result["accuracy"]
      best_config["random"] = best
```

So, we searched the hyperparameter space. Problem is, these spaces potentially can get pretty big. Let's imagine you have 10 hyperparameters, and every hyperparameter has 5 possible (relevant) values, you already have 5^{10} possible combinations, which is almost 10 million. Even if checking of every configuration would take just 1 second, it would take more than a 100 days to check them all...This space can grow out of control pretty fast.

Lets look at the best results we found.

```
[17]: best
```

```
[17]: {'input_size': 3,
      'output_size': 20,
      'dropout': 0.05,
      'epochs': 10,
      'hidden_size': 293,
```

```

'num_layers': 3,
'tune_dir': PosixPath('/Users/nickreinders/Documents/Master_Applied_Data_Scienc
e_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/notebooks/4_tuning_networks/logs/test'),
'accuracy': 0.996875}

```

```

[18]: columns = ["config/hidden_size", "config/num_layers", "accuracy"]
visualize.parallel_plot(analysis, columns)

```

Note how the mean scores are sort of randomly distributed. This is a direct effect of random guessing parameters.

3 Grid search

We can do this more rigorous with a gridsearch. The upside of this technique is that you will test every configuration.

A huge downside is the inefficiency. Also, you will run experiments with combinations that might be not very promising, but very slow, which is pretty inefficient.

One way to handle this is to use doubling as a strategy to scan the range: 16, 32, ..., 512. This is a bit more efficient.

Typically, this can be a good idea for a first scan, to get a rough idea of the space. After you have done this, you can narrow your searchspace, and do a more fine grained search zoomed in on areas that seem promising.

```

[19]: config = {
    "input_size": 3,
    "output_size": 20,
    "dropout": 0.1,
    "epochs": MAX_EPOCHS,
    "hidden_size": tune.grid_search([16, 32, 64, 128, 256, 512]),
    "num_layers": tune.grid_search([2, 4, 8]),
    "tune_dir": tune_dir,
}

```

```

[20]: tic = time.time()

analysis = tune.run(
    tune_model,
    config=config,
    metric="valid_loss",
    mode="min",
    storage_path=str(tune_dir),
    stop={"training_iteration": MAX_EPOCHS},
    verbose=1,
)

```

```

timer["ray_grid"] = time.time() - tic

best = analysis.get_best_config()
best["accuracy"] = analysis.best_result["accuracy"]
best_config["grid"] = best

```

<IPython.core.display.HTML object>

```

2025-10-03 13:42:24,621 INFO tune.py:1009 -- Wrote the latest version of all
result files and experiment state to '/Users/nickreinders/Documents/Master_Appli
ed_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/notebooks/4_tuning_networks/logs/test/tune_model_2025-10-03_13-37-51' in
0.0074s.
2025-10-03 13:42:24,625 INFO tune.py:1041 -- Total run time: 273.35 seconds
(273.30 seconds for the tuning loop).
2025-10-03 13:42:24,625 INFO tune.py:1041 -- Total run time: 273.35 seconds
(273.30 seconds for the tuning loop).

```

```

[21]: import pandas as pd
all_results = pd.concat([all_results, analysis.results_df])

```

```

[22]: grid = analysis.results_df
plot_contour(grid, "config/hidden_size", "config/num_layers", "accuracy",
↪start=0.7)

```

As you can see, we get a systematic scan, but large parts of the space are still unexplored, and we also explored parts that are really not very promising.

```

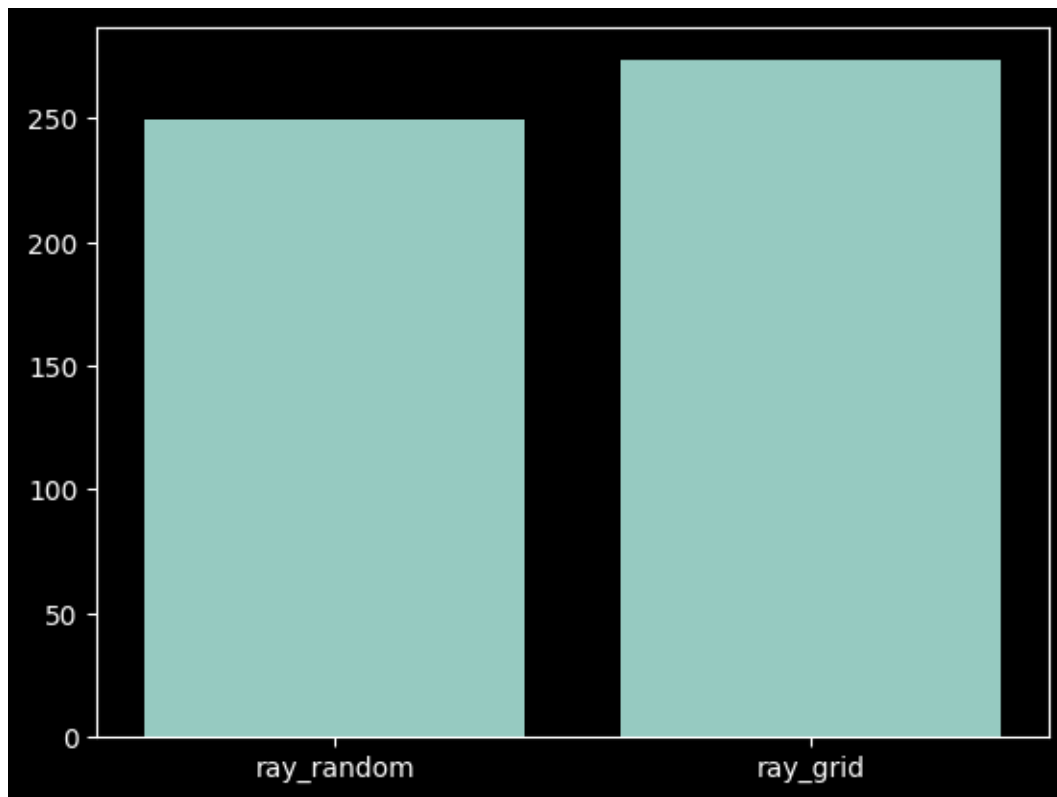
[23]: visualize.parallel_plot(analysis, columns)

```

```

[24]: visualize.plot_timers(timer)

```

4 Bayes

Now, we improve the search algorithm with a bayesian optimization.

Note that the bayesian search algorithm will only work with continuous parameters. This is a problem for the number of layers, which is a discrete parameter. I fixed this by simply casting the parameters to an integer inside the model, which is not very elegant but it works.

```
[25]: from ray.tune.search.bayesopt import BayesOptSearch

bayesopt = BayesOptSearch(metric="valid_loss", mode="min")

config = {
    "input_size": 3,
    "output_size": 20,
    "dropout": 0.1,
    "epochs": MAX_EPOCHS,
    "hidden_size": tune.uniform(16, 512),
    "num_layers": tune.uniform(1, 8),
    "tune_dir": tune_dir,
}
```

```

tic = time.time()

analysis = tune.run(
    tune_model,
    config=config,
    metric="valid_loss",
    mode="min",
    storage_path=str(tune_dir),
    num_samples=N_EXPERIMENTS,
    stop={"training_iteration": MAX_EPOCHS},
    search_alg=bayesopt,
    verbose=1,
)

timer["ray_bayes"] = time.time() - tic

best = analysis.get_best_config()
best["accuracy"] = analysis.best_result["accuracy"]
best_config["bayes"] = best

```

<IPython.core.display.HTML object>

```

2025-10-03 14:37:46,771 INFO bayesopt_search.py:293 -- Skipping duplicated
config: {'hidden_size': np.float64(229.49357578453206), 'num_layers':
np.float64(1.5357464325395038)}.
2025-10-03 14:37:54,648 INFO bayesopt_search.py:293 -- Skipping duplicated
config: {'hidden_size': np.float64(229.49357578453206), 'num_layers':
np.float64(1.5357464325395038)}.
2025-10-03 14:37:54,778 INFO bayesopt_search.py:293 -- Skipping duplicated
config: {'hidden_size': np.float64(229.49357578453206), 'num_layers':
np.float64(1.5357464325395038)}.
2025-10-03 14:39:03,994 INFO tune.py:1009 -- Wrote the latest version of all
result files and experiment state to '/Users/nickreinders/Documents/Master_Appli
ed_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/notebooks/4_tuning_networks/logs/test/tune_model_2025-10-03_14-35-48' in
0.0107s.
2025-10-03 14:39:03,997 INFO tune.py:1041 -- Total run time: 195.91 seconds
(195.89 seconds for the tuning loop).

```

[26]: best

```

[26]: {'input_size': 3,
      'output_size': 20,
      'dropout': 0.1,
      'epochs': 10,
      'hidden_size': np.float64(271.2563060950412),
      'num_layers': np.float64(3.173292899727462),

```

```
'tune_dir': PosixPath('/Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/notebooks/4_tuning_networks/logs/test'),
'accuracy': 0.9921875}
```

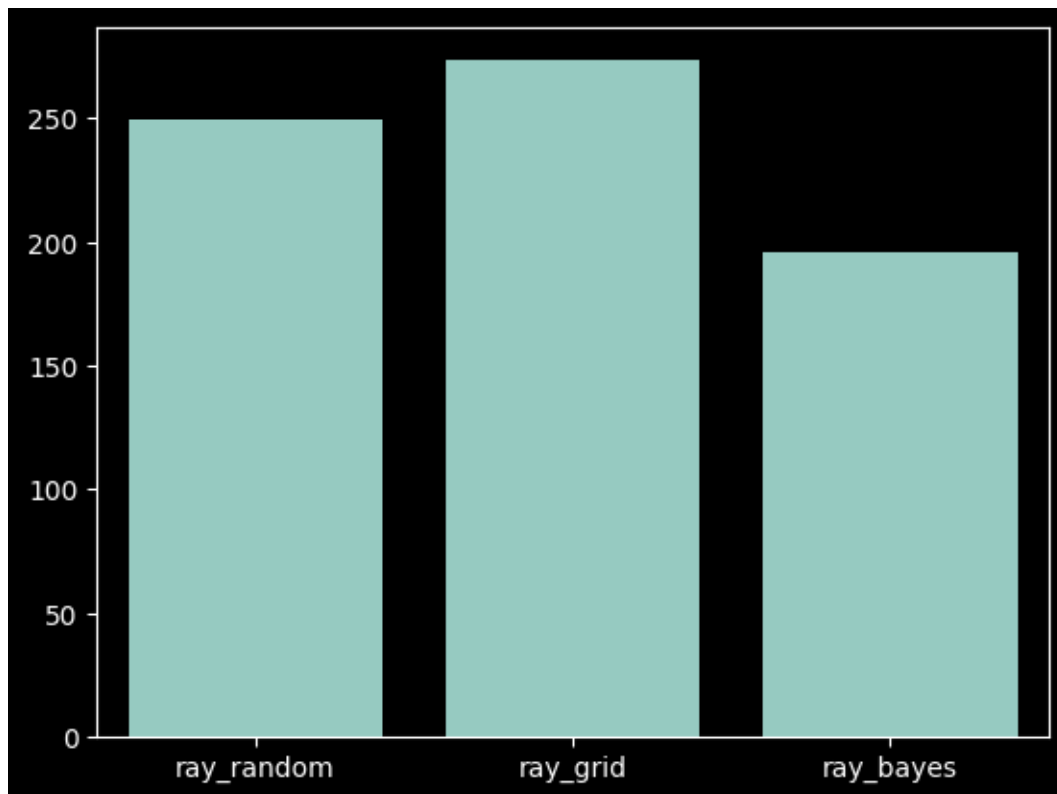
```
[27]: all_results = pd.concat([all_results, analysis.results_df])
```

```
[28]: bayes = analysis.results_df
plot_contour(bayes, "config/hidden_size", "config/num_layers", "accuracy",
↪start=0.7)
```

As you can see, bayes really focuses on the promising areas. It is much more efficient than random search, and also more efficient than grid search. We have set the `random_search_steps` to 5, this means that we will do 5 random searches first, to get a good initial idea of the space. As you can see, 5 is a bit low, because it might lead to premature converging to a local optimum. You can increase this number to get a better initial idea of the space, but you will also need to increase the number of iterations after the initial random scan.

```
[29]: visualize.parallel_plot(analysis, columns)
```

```
[30]: visualize.plot_timers(timer)
```



```
[31]: import pandas as pd
```

```
pd.DataFrame.from_dict(best_config, orient="index")
```

```
[31]:
```

	input_size	output_size	dropout	epochs	hidden_size	num_layers	\
random	3	20	0.05	10	293.000000	3.000000	
grid	3	20	0.10	10	512.000000	2.000000	
bayes	3	20	0.10	10	271.256306	3.173293	

	tune_dir	accuracy
random	/Users/nickreinders/Documents/Master_Applied_D...	0.996875
grid	/Users/nickreinders/Documents/Master_Applied_D...	0.993750
bayes	/Users/nickreinders/Documents/Master_Applied_D...	0.992188

5 Hyperband

Hyperband aborts runs early. Configs that are unpromising are abandoned before they complete.

```
[32]: from ray.tune.schedulers import AsyncHyperBandScheduler

scheduler = AsyncHyperBandScheduler(
    time_attr="training_iteration", grace_period=1, reduction_factor=3,
    ↪max_t=MAX_EPOCHS
)

config = {
    "input_size": 3,
    "output_size": 20,
    "dropout": 0.1,
    "epochs": MAX_EPOCHS,
    "hidden_size": tune.randint(16, 512),
    "num_layers": tune.randint(1, 8),
    "tune_dir": tune_dir,
}

tic = time.time()
analysis = tune.run(
    tune_model,
    config=config,
    metric="valid_loss",
    mode="min",
    storage_path=str(tune_dir),
    num_samples=N_EXPERIMENTS,
    stop={"training_iteration": MAX_EPOCHS},
    scheduler=scheduler,
    verbose=1,
)
```

```

timer["ray_hyperband"] = time.time() - tic

best = analysis.get_best_config()
best["accuracy"] = analysis.best_result["accuracy"]
best_config["hyperband"] = best

```

<IPython.core.display.HTML object>

```

2025-10-03 20:10:32,615 INFO tune.py:1009 -- Wrote the latest version of all
result files and experiment state to '/Users/nickreinders/Documents/Master_Appli
ed_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/notebooks/4_tuning_networks/logs/test/tune_model_2025-10-03_20-07-35' in
0.0101s.

```

```

2025-10-03 20:10:32,619 INFO tune.py:1041 -- Total run time: 176.94 seconds
(176.90 seconds for the tuning loop).

```

```

2025-10-03 20:10:32,619 INFO tune.py:1041 -- Total run time: 176.94 seconds
(176.90 seconds for the tuning loop).

```

If you study the iter column, you will see that some runs have been stopped early. This is exactly the point of hyperband: it tries to allocate resources to the most promising configurations.

The downside of this, is that if you try to get a good view of the space, you get distorted results because comparing a model that has trained just 1 or 3 epochs with a model that has trained 100 epochs is not very fair...

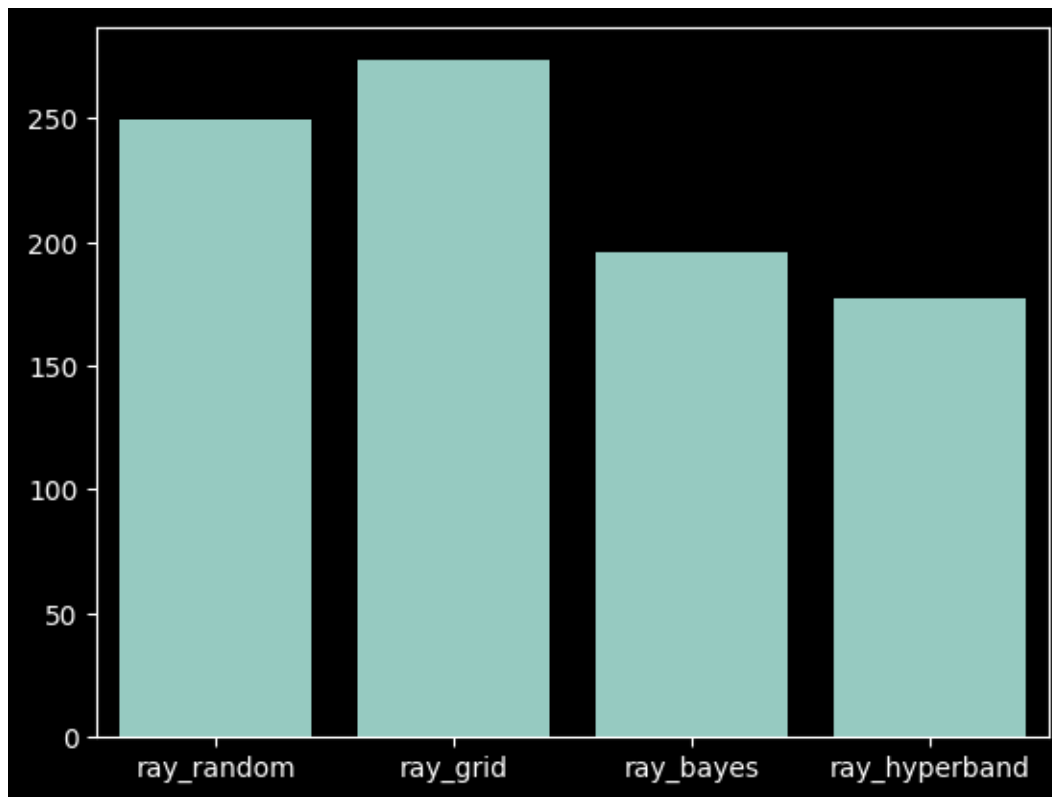
```
[33]: all_results = pd.concat([all_results, analysis.results_df])
```

```
[35]: hyperband = analysis.results_df
      plot_contour(hyperband, "config/hidden_size", "config/num_layers", "accuracy",
      ↪start=0.7)
```

```
[36]: visualize.parallel_plot(analysis, columns)
```

But note that it is faster! This means you could do more runs in the same time, which might be a good tradeoff! Typically you will get better results if you scan the searchspace better by doing more runs, aborting the ones that are not promising and going on with the ones that seem to yield good results.

```
[37]: visualize.plot_timers(timer)
```



```
[38]: pd.DataFrame.from_dict(best_config, orient="index")
```

```
[38]:
```

	input_size	output_size	dropout	epochs	hidden_size	num_layers	\
random	3	20	0.05	10	293.000000	3.000000	
grid	3	20	0.10	10	512.000000	2.000000	
bayes	3	20	0.10	10	271.256306	3.173293	
hyperband	3	20	0.10	10	466.000000	4.000000	

	tune_dir	accuracy
random	/Users/nickreinders/Documents/Master_Applied_D...	0.996875
grid	/Users/nickreinders/Documents/Master_Applied_D...	0.993750
bayes	/Users/nickreinders/Documents/Master_Applied_D...	0.992188
hyperband	/Users/nickreinders/Documents/Master_Applied_D...	0.993750

6 Hyperbayes

It is possible to combine Hyperband with Bayesian optimization, implemented as TuneBOHB in ray. However, TuneBOHB in ray is dependent on `hpbanner`, which is not maintained anymore. Unfortunately, a dependency of `hpbanner` is `netifaces`, which is also not maintained anymore. While `netifaces` does work on some hardware, it fails to build on some other hardware, which is a problem.

To still use BOHB type algorithms, I am planning to implement either SMAC3 or BEHB in the future.

7 HyperOpt

HyperOpt provides gradient/derivative-free optimization. They implemented TPE, which stands for Tree-structured Parzen Estimator. This is a bayesian optimization algorithm that uses a tree to model the distribution of the objective function. The main advantage is that TPE also works with discrete parameters and conditional search spaces which is a problem for the bayesian optimization.

For example, you could do things like this:

```
conditional_space = {
    "activation": hp.choice(
        "activation",
        [
            {"activation": "relu", "mult": hp.uniform("mult", 1, 2)},
            {"activation": "tanh"},
        ],
    ),
}
```

Where you have a discrete parameter `activation` that has a conditional parameter `mult`. This means that if you choose `relu`, you also need to set `mult`, but if you choose `tanh`, you do not need to set `mult`. See [documentation of ray](#) for more details.

```
[39]: from ray.tune.search.hyperopt import HyperOptSearch
search = HyperOptSearch()

scheduler = AsyncHyperBandScheduler(
    time_attr="training_iteration", grace_period=1, reduction_factor=3,
    ↪max_t=MAX_EPOCHS
)
```

```
[40]: config = {
    "input_size": 3,
    "output_size": 20,
    "dropout": 0.1,
    "epochs": MAX_EPOCHS,
    "hidden_size": tune.randint(16, 512),
    "num_layers": tune.randint(1, 8),
    "tune_dir": tune_dir,
}

tic = time.time()
analysis = tune.run(
    tune_model,
    config=config,
    metric="valid_loss",
```



```

mode="min",
storage_path=str(tune_dir),
num_samples=N_EXPERIMENTS,
stop={"training_iteration": MAX_EPOCHS},
search_alg=search,
scheduler=scheduler,
verbose=1,
)

timer["ray_hyopt"] = time.time() - tic

best = analysis.get_best_config()
best["accuracy"] = analysis.best_result["accuracy"]
best_config["hyopt"] = best

```

<IPython.core.display.HTML object>

```

2025-10-04 08:20:30,116 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}
2025-10-04 08:20:35,372 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}
2025-10-04 08:20:40,799 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}
2025-10-04 08:20:49,515 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}
2025-10-04 08:20:54,219 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}
2025-10-04 08:20:56,784 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}
2025-10-04 08:21:01,052 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}
2025-10-04 08:21:03,791 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}
2025-10-04 08:21:05,161 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}
2025-10-04 08:21:11,258 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}
2025-10-04 08:22:14,015 INFO tensorboardx.py:308 -- Removed the following

```

```

hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}]
2025-10-04 08:22:31,602 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}]
2025-10-04 08:23:02,727 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}]
2025-10-04 08:23:07,118 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}]
2025-10-04 08:23:35,438 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}]
2025-10-04 08:23:51,044 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}]
2025-10-04 08:23:54,010 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}]
2025-10-04 08:24:25,844 INFO tensorboardx.py:308 -- Removed the following
hyperparameter values when logging to tensorboard: {'tune_dir': ('__ref_ph',
'b1421250')}]
2025-10-04 08:24:25,853 INFO tune.py:1009 -- Wrote the latest version of all
result files and experiment state to '/Users/nickreinders/Documents/Master_Appli
ed_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-
course/notebooks/4_tuning_networks/logs/test/tune_model_2025-10-04_08-20-22' in
0.0078s.
2025-10-04 08:24:25,857 INFO tune.py:1041 -- Total run time: 243.23 seconds
(243.20 seconds for the tuning loop).

```

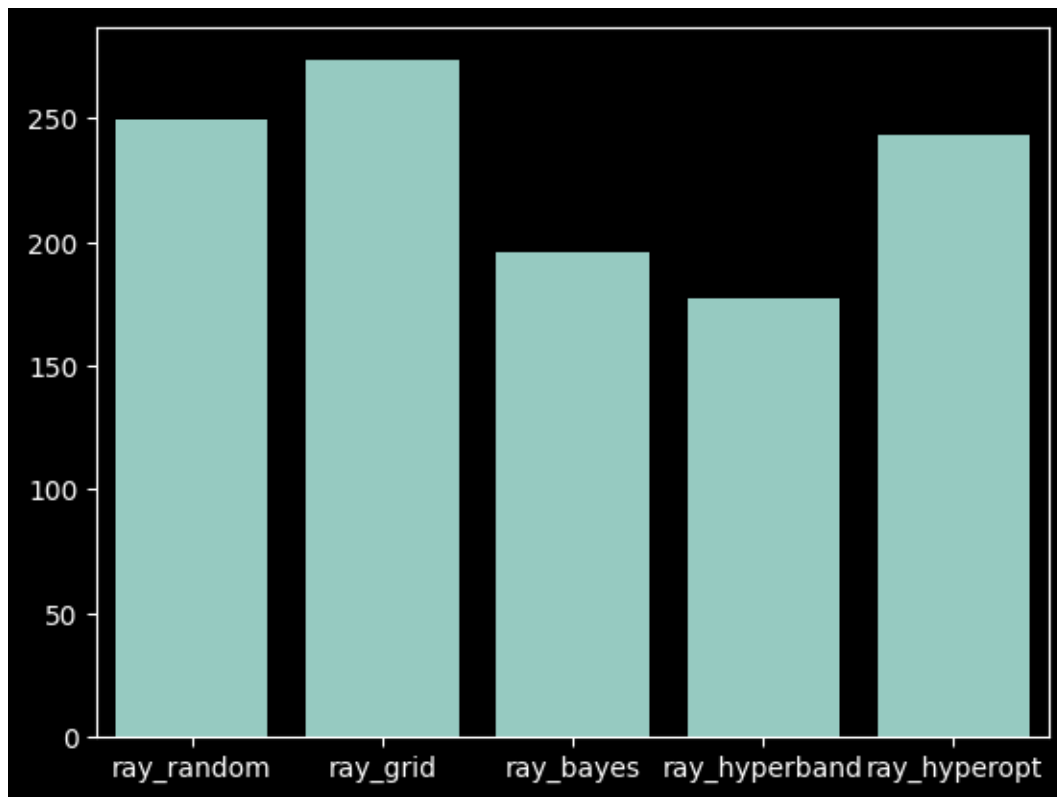
```
[41]: all_results = pd.concat([all_results, analysis.results_df])
```

```
[42]: hyperbayes = analysis.results_df
      plot_contour(hyperbayes, "config/hidden_size", "config/num_layers", "accuracy",
      ↪start=0.3, size=0.05)
```

This model combines bayesian optimization with hyperband. This is a good idea, because it combines the efficiency of bayesian optimization with the speed of hyperband.

```
[43]: visualize.parallel_plot(analysis, columns)
```

```
[44]: visualize.plot_timers(timer)
```



```
[45]: pd.DataFrame.from_dict(best_config, orient="index")
```

```
[45]:
```

	input_size	output_size	dropout	epochs	hidden_size	num_layers	\
random	3	20	0.05	10	293.000000	3.000000	
grid	3	20	0.10	10	512.000000	2.000000	
bayes	3	20	0.10	10	271.256306	3.173293	
hyperband	3	20	0.10	10	466.000000	4.000000	
hyperopt	3	20	0.10	10	388.000000	3.000000	

	tune_dir	accuracy
random	/Users/nickreinders/Documents/Master_Applied_D...	0.996875
grid	/Users/nickreinders/Documents/Master_Applied_D...	0.993750
bayes	/Users/nickreinders/Documents/Master_Applied_D...	0.992188
hyperband	/Users/nickreinders/Documents/Master_Applied_D...	0.993750
hyperopt	/Users/nickreinders/Documents/Master_Applied_D...	0.987500

```
[46]: contour = all_results[all_results["training_iteration"] == MAX_EPOCHS]
plot_contour(contour, "config/hidden_size", "config/num_layers", "accuracy",
↪start=0.8)
```

```
[47]: stop = time.time() - start
print(stop)
```

209902.87967514992

```
[48]: if DELETE:
      import shutil
      shutil.rmtree(tune_dir)
```